ModelIt!

Teacher's Guide

How Do We Take Care of Earth?

Reduce, Reuse, and Recycle

Quick Reference

Grade Level	Kindergarten
NGSS Standard	K-ESS3-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: K-ESS3-3

Communicate solutions that will reduce the impact of humans on the land, water, air, and/or other living things in the local environment.

Three-Dimensional Learning

Science Practice	**Obtaining, Evaluating, and Communicating Information** — Students gather information about how trash affects the environment and communicate solutions like reducing, reusing, and recycling.
Disciplinary Core Idea	**ESS3.C Human Impacts on Earth Systems** — Things that people do to live comfortably can affect the world around them. But they can make choices that reduce their impacts on the land, water, air, and other living things.
Crosscutting Concept	**Cause and Effect** — Students observe that more trash (cause) leads to more pollution (effect), and that reducing trash reverses this relationship.

CAST Testing Alignment

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Humans produce billions of tons of trash every year, much of which ends up in landfills, oceans, and natural habitats. Teaching children about reducing waste empowers them to become responsible stewards of the environment from an early age.

How Trash Affects the Earth

When people throw things away, the trash goes to landfills — huge areas of land filled with garbage. Some trash ends up in rivers and oceans, where it can harm fish, sea turtles, birds, and other wildlife. Animals can eat plastic thinking it is food, or get tangled in trash. Chemicals from garbage can leak into soil and groundwater, making water unsafe to drink. The more trash humans produce, the more pollution and habitat destruction result.

Reduce, Reuse, Recycle

The three R's provide a framework for minimizing waste. Reduce means using less in the first place — choosing a reusable water bottle instead of buying plastic ones. Reuse means finding new purposes for items instead of throwing them away — turning a cardboard box into a toy castle. Recycle means sending materials like paper, plastic, glass, and metal to facilities that process them into new products. Reducing is the most effective strategy because it prevents trash from being created at all. Even small actions by young children — like using both sides of paper — add up to meaningful impact.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Amount of Trash as the external component (people decide how much they throw away) and Pollution Level as the internal component (it changes based on how much trash ends up in the environment).

E **Establish Relationships**
Students discover that more trash leads to more Pollution Level — a positive relationship.

V **Visualize & Model**
Students build a simple two-part model connecting Amount of Trash to Pollution Level with a positive arrow.

E **Evaluate Outcomes**
Students test Lots of Trash vs. Less Trash scenarios and observe the difference in how clean or dirty the park looks.

R **Revise & Extend**
Students explore what happens when they add recycling — if trash goes into recycling bins instead of the ground, does pollution go down?

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Look at this beautiful park! Now look at THIS park with trash everywhere. Which one would you rather play in? Today we learn how to keep our Earth clean!"*

Watch For

Show two pictures: a clean park and a littered park. Ask: What is different? How did the trash get there? Can we fix it?

Slide 2: What Will We Learn?

3 min

Say: *"Today we learn THREE magic words to help the Earth: REDUCE — use less. REUSE — use it again. RECYCLE — make it into something new!"*

Watch For

Teach the three R's with gestures: Reduce (make yourself small), Reuse (spin around — use it again!), Recycle (make a circle with your arms). Practice all three.

Slide 3: Let's Be Scientists!

3 min

Say: *"Our big question is: What can WE do to help take care of our planet? Even little kids can make a BIG difference!"*

Watch For

Pair-share: Tell your partner one way you could make less trash at home or school. Share ideas with the class.

Slide 4: What Are the Parts?

5 min

Say: *"Our model has two parts. Amount of Trash is how much garbage people throw away — WE decide that! Pollution Level is how dirty the Earth gets — that depends on our trash."*

Watch For

Use sorting mats. Students place Amount of Trash on OUTSIDE and Pollution Level on INSIDE. Show a clean rug, then scatter paper scraps on it — that is what trash does to the Earth!

Slide 5: How Are They Connected?

5 min

Say: *"When we throw away MORE trash, Pollution goes UP — the Earth gets dirtier. That is a POSITIVE connection. But if we make LESS trash, pollution goes DOWN!"*

Watch For

Demonstrate: students scatter paper scraps (trash) on a mat (Earth). More trash = dirtier. Now pick it up and recycle it. Less trash = cleaner! Thumbs up for positive.

Slide 6: What Did We Find Out?

5 min

Say: *"We found out that WE can help the Earth! Reduce, Reuse, Recycle — three ways to make less trash and keep our planet beautiful."*

Watch For

Chant: 'When Trash goes up, Pollution goes up!' But then: 'When we REDUCE, pollution goes DOWN!' Complete the sentence frame together.

Slide 7: STEM Challenge

10 min

Say: *"Time for the Reuse Art Show! Take something from the recycling bin and turn it into something AMAZING. Trash into treasure!"*

Watch For

Set up a reuse station with clean recycled materials: boxes, bottles, tubes, newspapers, tape, glue. Students create a new useful or fun item. Display in a class art show and explain what they reused.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Amount of Trash
- **INTERNAL:** Pollution Level

Relationships

- **Amount of Trash to Pollution Level:** POSITIVE (+) — When people throw away more trash, pollution increases because the garbage piles up on land, washes into water, and hurts animals and their habitats.

Key Discoveries

- More trash makes more pollution — dirty water, dirty land, and hurt animals
- When we reduce, reuse, and recycle, we make LESS trash and the Earth stays cleaner
- Even kids can help by using less stuff, reusing containers, and recycling paper and plastic

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Trash just disappears when the garbage truck takes it"	Trash does not disappear — the garbage truck takes it to a landfill where it piles up in a huge area. Some trash takes hundreds of years to break down. Plastic bottles can last 450 years! That is why reducing trash is so important.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Recycling is the same as throwing things away"	Recycling is very different from throwing things in the trash. When you recycle, the materials are collected, cleaned, and turned into brand new things. A plastic bottle can become a new bottle or even a t-shirt! When you throw things in the trash, they sit in a landfill forever.